Unemployment Trends in India

Regional and monthly analysis with a descriptive COVID-era comparison

PROJECT SUMMARY

This project explores regional and temporal patterns in India unemployment survey estimates using the supplied area-level CSV.

DATA AND SCOPE

The primary area-level file contains 768 rows before cleaning and 740 after date parsing, missing-key removal, and duplicate removal. The observed date range is May 2019 to June 2020. A second state-level CSV is not pooled because it has a different grain.

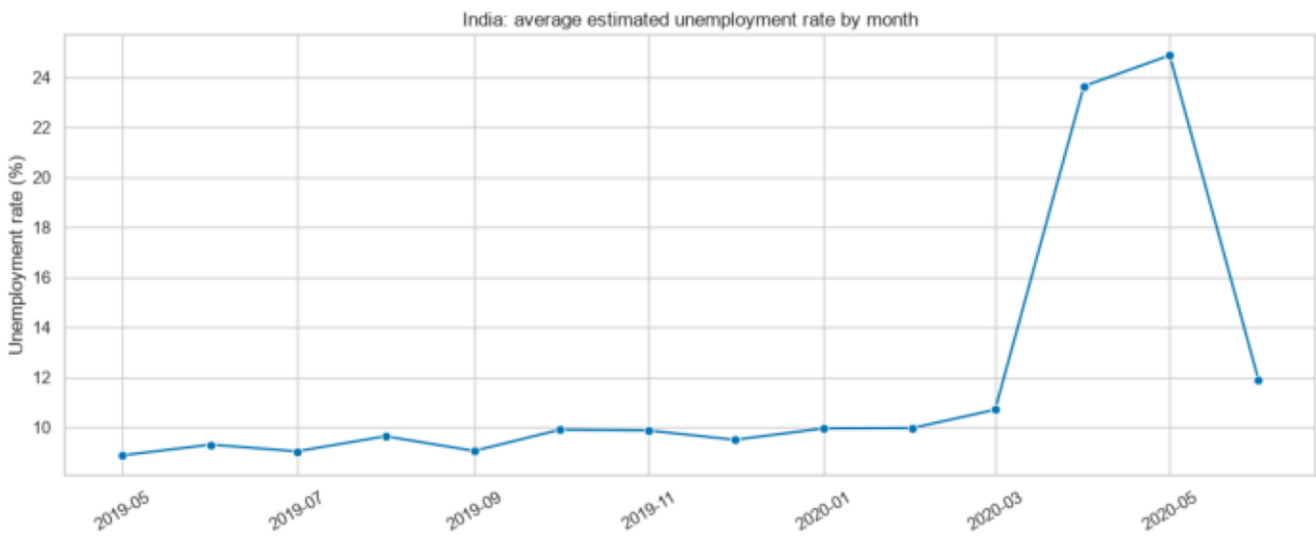
ANALYSIS APPROACH

Converted date and numeric fields; inspected shape, types, missingness, and descriptive statistics; calculated regional averages; plotted monthly trends, three well-covered state trajectories, and the top ten regional averages; and examined correlations among unemployment, employment, and labour participation estimates. Compared periods before March 2020 and from March 2020 onward.

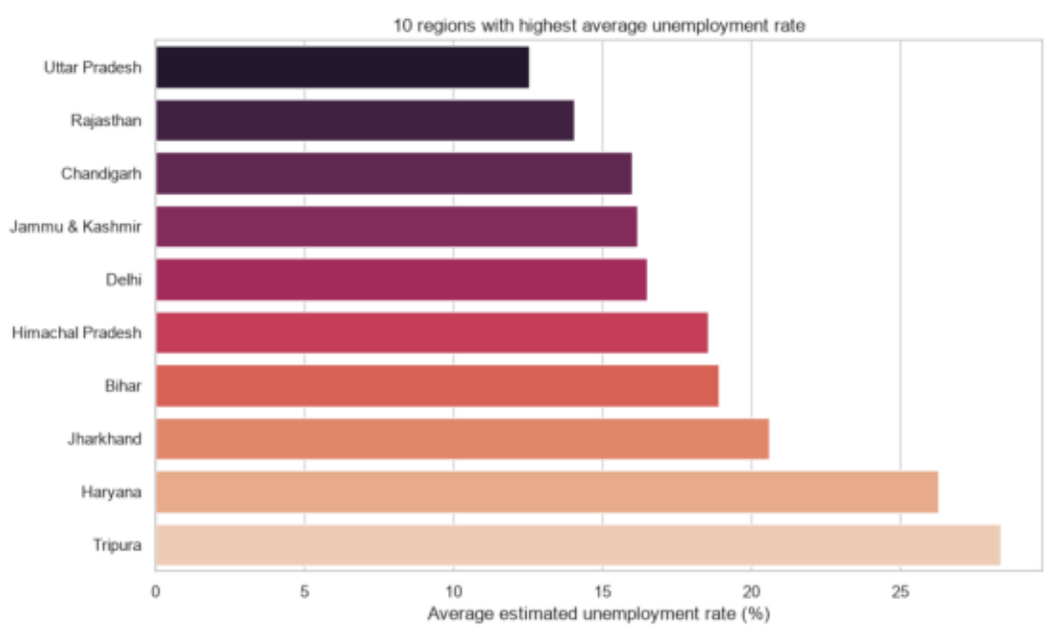
Prepared from the executed project notebook and supplied data.

RESULTS AND EVIDENCE

The area-weighted mean unemployment estimate was 9.51% before March 2020 and 17.77% from March 2020 onward in this dataset. Tripura had the highest observed region-level mean (28.35%). The post-onset rise is descriptive and cannot isolate a pandemic effect from seasonality, other events, or survey measurement.



Selected notebook visualization



Selected notebook visualization

INTERPRETATION AND REPRODUCTION

Limitations and responsible use

The available period is short and ends in June 2020. Area observations are equally weighted in regional averages, and the pre/post windows differ in length. The comparison is not causal evidence.

Reproduce the analysis

Open unemployment_india_eda.ipynb in Jupyter and run all cells. Keep Unemployment in India.csv in this folder.

The notebook includes the complete EDA, charts, modeling code, metric tables, and cell outputs.

Interpret model scores in the context of the dataset and holdout design. Observational patterns and predictive associations do not establish causal effects.